Band-1 Finite Ledger: Dedekind numbers M(n) — known exact values, box recomputation, next-n frontier

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Band-1 Finite Ledger: Dedekind numbers $M(n)$ (no large-n closure)

Honesty stamp. This note does **not** claim that Dedekind's problem is solved for large n. It does **not** stamp Clay-closed, finished ToE, or any prize claim. It records (i) the published exact ledger $M(0), \dots, M(9)$ with sources, (ii) an independent box recomputation of $M(n)$ for $n \leq 6$, and (iii) the next open computational frontier $M(10)$.

Abstract

The Dedekind number $M(n)$ counts monotone Boolean functions on n variables (equivalently: antichains in the Boolean lattice $2^{[n]}$, or downsets of that lattice). Exact values are known only through $n = 9$ (as of 2023–2024). On this Linux box we **independently recompute** $M(0), \dots, M(6)$ by two methods (downset DFS; MBF pair recursion) and match the literature bit-for-bit. Full on-box certification of $M(7)$ via naive $O(|\mathrm{MBF}_6|^2)$ pair enumeration was **not** completed (many CPU-hours); $M(7)$–$M(9)$ are cited from the published record only. **Next open exact n: 10.**

Statement locked

Definition. Let $M(n)$ be the number of monotone Boolean functions $f : \{0,1\}^n \to \{0,1\}$ (i.e., $x \leq y$ coordinatewise $\Rightarrow f(x) \leq f(y)$). Equivalent counts:

- number of antichains in $(2^{[n]}, \subseteq)$ (including $\emptyset$);
- number of downsets (order ideals) of $(2^{[n]}, \subseteq)$;
- number of elements of the free distributive lattice on n generators, plus the empty/top conventions that yield OEIS A000372.

Classical recursion. Every monotone f on n variables decomposes uniquely as a comparable pair (g, h) of monotone functions on $n-1$ variables with $g \leq h$ pointwise, via

$$f(x_1, \dots, x_{n-1}, x_n) = (\neg x_n \wedge g(x')) \vee (x_n \wedge h(x')).$$

Hence $M(n) = \#\{(g,h) : g, h \text{ monotone on } n-1,\ g \leq h\}$.

Literature ledger (published exact values)

Exact $M(n)$ are known for $0 \leq n \leq 9$ (OEIS A000372; Wikipedia *Dedekind number*; MathWorld). Discoverers:

n	$M(n)$	First computation (source)
0	2	Dedekind (1897)
1	3	Dedekind (1897)
2	6	Dedekind (1897)
3	20	Dedekind (1897)
4	168	Dedekind (1897)
5	7581	Church (1940)
6	7828354	Ward (1946)
7	2414682040998	Church (1965)
8	56130437228687557907788	Wiedemann (1991)
9	286386577668298411128469151667598498812366	Jäkel (2023); Van Hirtum et al. (2024), independent

Sources (statement + values).

- OEIS A000372: https://oeis.org/A000372
- Wikipedia, *Dedekind number* (accessed 2026-09-11): https://en.wikipedia.org/wiki/Dedekind_number
- MathWorld, *Dedekind Number*: https://mathworld.wolfram.com/DedekindNumber.html
- R. Dedekind, *Über Zerlegungen von Zahlen durch ihre grössten gemeinsamen Teiler*, Gesammelte Werke Bd. 1 (1897).
- R. Church, *Numerical analysis of certain free distributive structures*, Duke Math. J. **6** (1940); and Notices AMS **12** (1965) for $M(7)$.
- M. Ward, *Note on the order of the free distributive lattice*, Bull. Amer. Math. Soc. **52** (1946).
- D. Wiedemann, *A computation of the eighth Dedekind number*, Order **8** (1991).
- C. Jäkel, *A computation of the ninth Dedekind number*, J. Computational Algebra **6–7** (2023) 100006; arXiv:2304.00895.
- L. Van Hirtum et al., *A computation of the ninth Dedekind number using FPGA supercomputing*, ACM TRETS **17** (2024); arXiv:2304.03039.

$M(9)$ was obtained independently by GPU matrix methods (Jäkel) and FPGA acceleration on Noctua 2 (Van Hirtum et al.); the two agree.

Finite computational lemma (only lemma in this note)

> **Lemma (finite, computational).** For each $n \in \{0,1,2,3,4,5,6\}$, the downset-DFS procedure of §Method returns exactly the integer $M(n)$ recorded in the literature table above. Independently, the MBF pair-recursion procedure builds $|\mathrm{MBF}_n| = M(n)$ for $n \le 6$ and confirms $M(6) = \#\{(g,h) : g \le h \text{ in } \mathrm{MBF}_5\}$.

Proof idea (machine-checkable). Process elements of $2^{[n]}$ in rank (popcount) order. At set S, exclusion is always legal; inclusion is legal iff every immediate subset is already chosen. Leaves of this binary tree are exactly the downsets; their count is $M(n)$. The pair recursion enumerates truth-table masks closed under the $g \le h$ glue and matches sizes. □

This lemma is **finite**: it does not determine $M(n)$ for $n \ge 7$ on this box, and it does not close Dedekind's problem.

Method

1. **Downset DFS (Method A).** Boolean lattice on 2^n elements; iterative stack DFS; Python arbitrary-width ints for the inclusion bitset. Used for $n \leq 6$.
2. **MBF pair recursion (Method B).** Build all monotone truth-table masks for $n-1$, retain pairs with $(g \,\&\, \neg h) = 0$, concatenate halves. Cross-check sizes through $n = 6$; obtain $M(6)$ as the comparable-pair count on MBF_5 (7581^2 scans, ~3 s).
3. **Not completed here.** Naive $O(M(6)^2) \approx 6 \cdot 10^{13}$ scans for $M(7)$; laptop-class wall-clock estimated in many CPU-hours without a specialized interval/FPGA/GPU pipeline.

Method limits

- Exact integer counts; no floating-point predicate.
- Box certification stops at $n = 6$ in this run.
- Literature values for $n = 7, 8, 9$ are **cited**, not re-derived on this box.
- **Explicitly not claimed / not done:** closed form for $M(n)$; exact $M(10)$; Clay / ToE / prize stamp; asymptotic Korshunov-type formulae as a substitute for exact $M(10)$.

Results (numbers!)

Table 1 — Box-certified $M(n)$ for $n \leq 6$ (Method A)

n	$M(n)$ (box)	Literature	Match	Wall (s)
0	**2**	2	yes	$< 10^{-4}$
1	**3**	3	yes	$< 10^{-4}$
2	**6**	6	yes	$< 10^{-4}$
3	**20**	20	yes	$< 10^{-4}$
4	**168**	168	yes	0.0002
5	**7581**	7581	yes	0.022
6	**7828354**	7828354	yes	24.6

Table 2 — Method B cross-check (MBF list sizes + $M(6)$ from pairs)

n	$\|\text{MBF}_n\|$ (box)	Literature $M(n)$	Match
0	2	2	yes
1	3	3	yes
2	6	6	yes
3	20	20	yes
4	168	168	yes
5	7581	7581	yes
6	7828354	7828354	yes

$M(6)$ via comparable pairs on MBF_5: **7828354** (match), ≈ 3.2 s pair scan; MBF_6 list build ≈ 3.9 s.

Table 3 — Literature-only $M(7)$–$M(9)$ + frontier

n	Status on this box	$M(n)$	Source
7	literature only (not re-enumerated)	2414682040998	Church (1965)
8	literature only	56130437228687557907788	Wiedemann (1991)
9	literature only (two independent 2023/24 computations)	286386577668298411128469151667598498812366	Jäkel; Van Hirtum et al.
10	**open**	**unknown**	—

Next open computational frontier

Exact $M(10)$ **is open.** $M(9)$ required GPU-hours (Jäkel) or ~47{,}000 FPGA-hours on Noctua 2 (Van Hirtum et al.). No credible claim of $M(10)$ appears in the published record as of this note's sources. Asymptotic estimates exist; they do **not** replace an exact integer.

On-box **next-n lab target** (if continuing this thread): specialized interval / permutation-reduced / matrix-product pipeline aimed at verifying $M(7)$ first (Church's value), then $M(8)$, before any attempt at $M(10)$.

Reproducibility

- Code: `/workspace/mission-control/labs/dedekind/recompute_dedekind.py`
- Results JSON: `/workspace/mission-control/labs/dedekind/results.json`
- Sibling copy: `/workspace/experiments/band1/dedekind/results.json`

Rerun: `python3 /workspace/mission-control/labs/dedekind/recompute_dedekind.py`

Conclusion

Box-certified: $M(0), \ldots, M(6) = (2, 3, 6, 20, 168, 7581, 7828354)$ by two independent algorithms, matching OEIS A000372. **Literature ledger** through $M(9)$ with classical and 2023/24 sources. **Frontier:** exact $M(10)$ open. No large-n closure; no Clay / ToE stamp.